
Swarm-Based Topology-Aware Optimization for Automatic Mapper Parameter Selection

Abstract: In the Mapper algorithm, the selection of input parameters, especially the resolution parameters, is still a big challenge. If the parameters are selected inappropriately, the quality of the resulting topological graphs can be significantly affected. This study proposes S-Mapper, a topological black-box optimization method that can automatically select resolution parameters for Mapper. This method considers the number of intervals and overlap rates as variables in the search space and optimizes them by using the topological recognition target function. This objective function is specified by the Topological Silhouette Coefficient (TSC). Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO) are used as typical optimization strategies to concretize this framework. Experiments on datasets including composite point cloud data, 3D grid data and actual data all show that S-Mapper generally produces more stable Mapper graphs with improved topological fidelity. In a specific experiment, S-Mapper achieved a TSC of 0.830 on the Horse dataset and 0.540 on the COVID-19 dataset, which is superior to other strategies. The experimental results show that the effectiveness of automatically optimizing parameter selection brings better results than manually adjusting parameter selection.

Keywords: Topological data analysis; Mapper algorithm; Topology-aware optimization; Automatic parameter selection; Black-box optimization

1 Introduction

Topological Data Analysis (TDA) has emerged as a powerful framework for understanding complex datasets through their intrinsic shape Akingbade et al. (2024); Pham et al. (2025). Instead of relying on linearity or distributional assumptions as in conventional statistical methods, TDA provides tools for capturing topological structures such as loops, branches, voids and connected components Pham (2025). Thanks to the ability of topological features to remain stable when encountering noisy and continuously distorted data, TDA is especially effective in applications such as biology, material science, as well as Fitzpatrick et al. (2025) signal processing. At the same time, TDA also provides a view to support analysis based on geometry or features thanks to the ability to focus on the shape of the data. As data complexity continues to increase, shape-oriented representations become extremely important for interpretation. Currently, TDA has also been successfully applied in many different fields, including cancer subtype classification, protein folding analysis, and other studies such as neural networks. These applications help highlight the robustness and uniqueness of TDA in extracting structural insights from heterogeneous data.

Among TDA tools, the Mapper algorithm is particularly useful for producing graph-based summaries of high-dimensional data. The algorithm constructs a simplicial graph

by applying a filter function to the data, covering the function’s range with overlapping intervals, and performing clustering within each corresponding preimage. With the obtained graphs, the necessary topological and geometric information has been encoded, helping to allow structural analysis of complex datasets. Thanks to that, parameterization and compatibility with many choices of filter functions and clustering methods are possible. Mapper is also refined for many data types and more diverse analytical objectives in different fields. Furthermore, the structural summaries have been integrated into downstream analytical pipelines for tasks such as classification, anomaly detection, and subgroup discovery tasks.

The Mapper algorithm is sensitive to resolution parameters, particularly the number of intervals and the overlap ratio in the filter space Singh et al. (2007). Inappropriate parameter settings may oversimplify or fragment Mapper graphs, making important topological patterns difficult to detect. Since parameter tuning is often heuristic and data-dependent, it limits the automation and scalability of Mapper analysis. To address this issue, prior studies have explored solutions such as rule-based interval partitioning combined with fuzzy clustering Bui et al. (2020), ensemble-based parameter fusion Kang and Lim (2021), information-theoretic cover adaptation via X-means Chalapathi et al. (2021), statistical refinement using G-means Alvarado et al. (2025), as well as normality testing Vejdemo-Johansson and Leshchenko (2020); thus, probabilistic modeling with stochastic optimization Tao and Ge (2025b), and distribution-guided cover construction based on intrinsic data characteristics Tao and Ge (2025a). This paper addresses the challenge of Mapper resolution parameter selection by introducing S-Mapper, a topology-aware black-box optimization framework for adaptive Mapper construction. The proposed framework treats the number of intervals and the overlap percentage as learnable resolution variables that are automatically selected from data rather than manually specified by users. A topology-aware quality signal is used as the optimization objective to guide the search toward stable and informative Mapper graphs. In this formulation, swarm metaheuristics serve only as concrete optimizers of the objective and are interchangeable. Crucially, the framework is also agnostic to the specific topological quality signal used as the objective: the Topological Silhouette Coefficient (TSC) serves as one concrete instantiation in this paper, while the framework is designed to allow other topology-aware quality measures to be substituted in its place. The contribution therefore lies in the topology-aware optimization formulation itself rather than in TSC or in any particular optimizer. We selected the PSO and GWO algorithms to illustrate the workings of the proposed theoretical framework. These are two typical optimization algorithms of the swarm optimization algorithm group; they use mechanically different update strategies, facilitating easy comparison of convergence behavior and graph quality obtained during optimization.

- (1) We introduce a novel framework, S-Mapper, which integrates swarm intelligence techniques to automate resolution parameter selection in the Mapper algorithm, thereby eliminating reliance on heuristic or manual tuning.
- (2) We demonstrate the effectiveness of swarm-based parameter selection through extensive experiments on diverse datasets—including synthetic point clouds, 3D meshes, and real-world biological data—using metrics that evaluate both clustering quality and topological fidelity.
- (3) We provide a comparative evaluation of two swarm-based strategies (PSO and GWO), highlighting their strengths, convergence behavior, and performance trade-offs in improving the quality and stability of Mapper outputs across varying data domains.

The remainder of this paper is organized as follows: Section 2 reviews relevant research on Mapper. Section 3 introduces the foundational knowledge of Mapper and the swarm intelligence techniques used in this study. Section 4 presents the proposed S-Mapper framework. Section 5 presents experimental results, comparisons, and performance evaluations of different parameter selection strategies. Section 6 concludes the paper and suggests future directions for development.

2 Related Work

In recent years, the Mapper algorithm has emerged as a foundation of TDA, a powerful and flexible method to extract structural information from complex and multidimensional datasets Chazal and Michel (2021), Smith et al. (2021). By transforming data into graph-based summaries constructed from overlapping intervals and local clustering, Mapper enables the identification of meaningful topological patterns Bui et al. (2021). Different from classical statistical methods that are only based on linear assumptions, Mapper helps capture nonlinear relationships and global structures of Carlsson (2020) data. Thanks to that powerful visual ability, Mapper has been successfully applied in many fields, demonstrating the ability to discover hidden patterns with minimal data preprocessing Singh et al. (2007). One of the most typical and earliest applications of Mapper is to help identify two separate subgroups of diabetic patients in Reaven and Miller’s diabetes dataset.

In biomedical applications, Mapper biology has shown a strong potential in finding important characteristics from noise and heterogeneous data Lum et al. (2013), Smith et al. (2024). It has been used to analyze the transformations related to the task and state of the brain to behavior Sagar et al. (2018). In cancer research, Mapper is used to identify new tumor suppressor genes Rabadán et al. (2020), Loughrey et al. (2021). In addition, Mapper also has other notable applications such as quantifying spatial patterns in stem cell clusters and classifying Alzheimer’s disease groups from RNA expression data Hartsock et al. (2025). These applications demonstrate the strong capability of Mapper to analyze multidimensional biological systems Geniesse et al. (2019).

Besides its role in data exploration, Mapper is also used as a functional module in graph-based deep learning models. Outstanding studies such as Deep Graph Mapper, by integrating Mapper as a combination mechanism in the graph neural network (GNN) it allows graph summary and topology recognition Bodnar et al. (2021). GraphPulse combines with Mapper with a regression neural network to model time graphs more effectively, improving the predictability of dynamic graph properties Shamsi et al. (2024). Meanwhile, Deep Mapper enhances molecular configuration modeling by simultaneously optimizing neural filters and clustering strategies on latent space created from auto encoders Oulhaj et al. (2024). These great advances have shown the importance of Mapper, it is not only a visualization or analysis tool but also a structural component in deep learning, learning topological representations.

Although the Mapper algorithm has proven to be effective in capturing meaningful topological structures and revealing intrinsic patterns in complex multidimensional data Carrière et al. (2018). But its performance depends greatly on the selection of input parameters, especially the resolution parameter including the number of coverage and the overlap rate, because these two parameters can directly affect the interpretability of the obtained graph. To overcome this weakness, some studies have proposed to reduce sensitivity and automate parameter selection such as F-Mapper, an extension based on

fuzzy clustering that automates the division of overlays by using the flexible adjustment of the overlap rate Bui et al. (2020). Following that success, Ensemble Mapper was born, by synthesizing the results of many Mapper outputs on different parameter settings to help graphs be represented more stably Kang and Lim (2021). Following that success, another study also uses the principle of graph synthesis proposed by Fitzpatrick et al. Fitzpatrick et al. (2023). This method helps to adjust all three main parameters of Mapper at the same time. It builds multiple Mapper graphs on diverse parameter configurations and selects optimal settings by maximizing consistency with the known label structure. Through extensive tests on labeled datasets, their combination-based strategy has affirmed its superiority while emphasizing its practical applicability in the real world.

Information theory and probability strategies are also used to be able to adjust the Mapper resolution automatically. Chalapathi et al. Chalapathi et al. (2021) introduced a method of creating adaptive coverage areas by using information criteria from the X-means clustering algorithm that allows the construction of dynamic intervals that are more suitable for the intrinsic data structure. In another related work, G-Mapper is also suggested that this is a combination of G-means clustering with Gaussian mixed models to help refine the coverage area appropriately. This technique has improved Mapper’s ability to retain structural characteristics while avoiding excessive division Alvarado et al. (2025). More recently, a probability frame-based method is called Soft Mapper, in which interval boundaries are modeled using a latent assignment matrix and optimized through the stochastic optimization Tao and Ge (2025b). And most recently, the D-Mapper proposed by Tao et al. Tao and Ge (2025a), this method allows for adjustment of overlap rates by leveraging probabilistic and robust homogeneous models. Therefore, this method is effective in capturing the characteristics of complex data such as viral genomes and revealing their evolutionary structure. These studies confirm that the selection and adjustment of Mapper parameters is essential and still has potential for further expansion.

3 Background

This section provides an overview of the fundamental concepts of Mappers, their evaluation methods, and the operating principles of optimization algorithms. Furthermore, Mappers and their underlying theoretical principles can be found in more detail in textbooks on TDA Carlsson and Vejdemo-Johansson (2021).

3.1 Mapper Algorithm

The Mapper algorithm first appeared in a study by Singh, Mémoli, and Carlsson in 2007 Singh et al. (2007). It subsequently became a foundational tool in TDA, used to represent and display the hidden shapes of complex and multidimensional datasets that traditional data analysis methods struggle to achieve. Based on topological structures, Mapper provides a system for detecting overall geometric structures such as connected components, loops, extensions, and holes that are often difficult to observe using linear or statistical projections Carrière and Oudot (2018).

Essentially, Mapper constructs a discrete approximation of the Reeb graph, representing the topological structure of a scalar function defined on a dataset. This process is performed step by step. First, the input data X is projected onto another space through a filter function $f : X \rightarrow \mathbb{R}$ where f can be a projection from techniques such as PCA or t-SNE or any

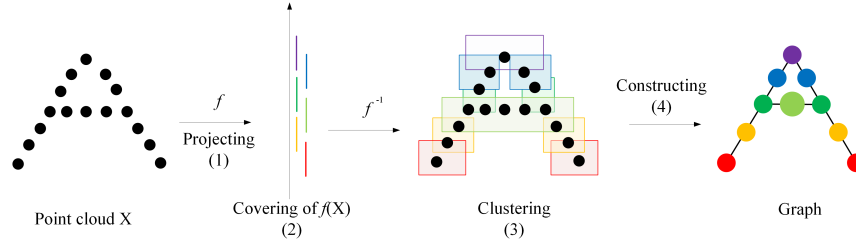


Figure 1: Step-by-step illustration of the Mapper algorithm applied to an A-shaped point cloud.

user-defined scalar function. The function f acts as a lens function to represent the dataset. Then, the image $f(X)$ is divided into N overlapping intervals $\{I_i\}_{i=1}^N$, where the overlap ratio $p \in (0, 1)$ controls the degree of intersection between adjacent intervals. The two parameters N and p are user-selected and are used to determine the resolution of the Mapper graph, thereby influencing the level of detail and connectivity of the resulting graph.

For each interval I_i , the inverse image $f^{-1}(I_i) \subset X$ is computed. The data points in each inverse image are further clustered using algorithms such as DBSCAN or K-means. Then, a Mapper graph is constructed with the number of nodes corresponding to the number of clusters, and two nodes are connected if the corresponding clusters have at least one common data point. The result is a graph structure that can reflect the geometric structure and density of the data. The entire process is illustrated in detail in Figure 1.

The expressive and interpretable capabilities of a Mapper graph depend heavily on input parameters, especially resolution parameters such as the number of intervals and overlap rate, because these parameters can directly alter the resulting graph. Inappropriate parameter selection can lead to a fragmented or overly simplified Mapper graph, making it difficult to detect important features. Therefore, selecting appropriate parameters is essential to balance resolution, stability, and graph representation.

3.2 Swarm Intelligence

SI is a class of computational techniques inspired by the collective behavior of decentralized, self-organized systems found in nature, such as flocks of birds, fish schools, or insect colonies. These biological systems exhibit emergent intelligence using simple local interactions among agents without the need for centralized control. SI algorithms model this phenomenon to solve complex optimization and search problems, especially in high-dimensional or nonlinear spaces. Characterized by adaptability, robustness, and scalability, SI has been widely adopted in engineering, data science, and artificial intelligence. Among the most established algorithms in this domain are PSO and GWO, both of which are population-based metaheuristics that simulate natural processes of movement, learning, and cooperation.

3.2.1 Particle Swarm Optimization

PSO is one of the classical algorithms for swarm optimization and has had a significant influence, introduced by Kennedy and Eberhart in 1995 Piotrowski et al. (2023). Through the collective behaviors of birds and fish, PSO considers each individual as a corresponding

candidate solution moving in the search space based on personal experience and information stored from neighboring individuals, thereby helping to balance exploration and exploitation during the optimization process Gong et al. (2024).

Mathematically, let $\mathbf{x}_i^t \in \mathbb{R}^d$ and $\mathbf{v}_i^t \in \mathbb{R}^d$ denote the position and velocity of the i -th particle at iteration t , respectively. The personal best position of the particle is denoted by $\mathbf{p}_i^t$, and the global best position found by the swarm is $\mathbf{g}^t$. Equations (1) and (2) govern the velocity and position updates:

$$\mathbf{v}_i^{t+1} = w\mathbf{v}_i^t + c_1r_1(\mathbf{p}_i^t - \mathbf{x}_i^t) + c_2r_2(\mathbf{g}^t - \mathbf{x}_i^t), \quad (1)$$

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1}, \quad (2)$$

here, w is the inertia weight controlling momentum, c_1 and c_2 are acceleration coefficients that weigh the cognitive and social components, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are random scalars promoting stochasticity. Over successive iterations, particles converge to regions of the search space that offer optimal or near-optimal values of the objective function.

3.2.2 Grey Wolf Optimizer

GWO, introduced by Mirjalili et al. in 2014, is a nature-inspired metaheuristic that emulates grey wolves' leadership hierarchy and collaborative hunting behavior in the wild Nadimi-Shahraki et al. (2024). In a typical grey wolf pack, the social hierarchy is divided into four groups: alpha (α), beta (β), delta (δ), and omega (ω). This hierarchical structure forms the foundation of the optimization mechanism in GWO, where the top three candidate solutions (α , β , and δ) guide the search of the remaining population (the ω -wolves). Grey wolves' encircling, hunting, and attacking processes are mathematically modeled to refine the population toward the global optimum iteratively Liu et al. (2024).

The position of a search agent (wolf) is denoted by $\mathbf{x}_i^t \in \mathbb{R}^d$ at iteration t . Let $\mathbf{x}_\alpha^t$, $\mathbf{x}_\beta^t$, and $\mathbf{x}_\delta^t$ be the top three positions in the population. The position of the i -th wolf is updated according to Equations (3):

$$\mathbf{x}_i^{t+1} = \frac{1}{3}(\mathbf{x}'_1 + \mathbf{x}'_2 + \mathbf{x}'_3), \quad (3)$$

where

$$\begin{aligned} \mathbf{x}'_1 &= \mathbf{x}_\alpha^t - \mathbf{A}_1 \cdot |\mathbf{C}_1 \cdot \mathbf{x}_\alpha^t - \mathbf{x}_i^t|, \mathbf{x}'_2 = \mathbf{x}_\beta^t - \mathbf{A}_2 \cdot |\mathbf{C}_2 \cdot \mathbf{x}_\beta^t - \mathbf{x}_i^t|, \\ \mathbf{x}'_3 &= \mathbf{x}_\delta^t - \mathbf{A}_3 \cdot |\mathbf{C}_3 \cdot \mathbf{x}_\delta^t - \mathbf{x}_i^t| \end{aligned}$$

In these expressions, $\mathbf{A}_k = 2\mathbf{a} \cdot \mathbf{r}_k - \mathbf{a}$ and $\mathbf{C}_k = 2 \cdot \mathbf{r}_k$, with $\mathbf{r}_k \sim \text{Uniform}(0, 1)$ and $\mathbf{a}$ decreasing linearly from 2 to 0 over the iterations. These components simulate the wolf pack's encircling mechanism and progressively intensify the search as the algorithm converges.

3.3 Evaluation

The Normalized Silhouette Coefficient (NSC) measures how well each data point fits within its assigned cluster while being separated from others Tao and Ge (2025a). Given a data point

x , let $a(x)$ denote the average intra-cluster distance and $b(x)$ the lowest average distance to any other cluster. The silhouette coefficient is defined as:

$$SC(x) = \frac{b(x) - a(x)}{\max\{a(x), b(x)\}}, \quad (4)$$

and the normalized form is:

$$NSC(x) = \frac{SC(x) + 1}{2}. \quad (5)$$

The NSC ranges from 0 to 1, where higher values indicate tighter clustering and better separability.

The Topological Signal Rate (TSR) quantifies the proportion of meaningful topological features in a Mapper graph by analyzing its extended persistence diagram Fasy et al. (2014). Using bottleneck bootstrap methods, points close to the diagonal are deemed noise, while those outside a confidence region are considered signals. TSR is calculated as:

$$TSR = \frac{N_{\text{signal}}}{M}, \quad (6)$$

where M is the total number of points and N_{signal} is the count of significant features. A higher TSR reflects a graph with a stronger and clearer topological structure.

The TSC, or Adjusted Silhouette Coefficient (SC_{adj}), integrates geometric clustering quality with topological fidelity Tao and Ge (2025a). It is computed by combining NSC and TSR through a weighted average:

$$TSC = w_1 \cdot NSC + w_2 \cdot TSR, \quad (7)$$

In this context, w_1 and w_2 allow us to decide whether the evaluation in the Mapper graph construction prioritizes cluster cohesion or topological signal preservation. Increasing w_1 prioritizes the evaluation of clustering quality by ignoring some topological signals; conversely, increasing w_2 focuses on evaluating whether the graph preserves topological signals while ignoring some clustering quality. Therefore, we decided to set the weights $w_1 = w_2 = 0.5$ to balance ensuring quality and preserving topological signals. We also performed a sensitivity analysis of these two weights when adjusting the objective function in the following section.

4 S-Mapper: A Topology-Aware Black-Box Optimization Framework for Adaptive Mapper Construction

In this section, we present the S-Mapper proposal framework, which handles the selection of the Mapper's resolution parameter as a topology-aware black-box optimization problem, allowing for automatic improvement of the stability and expressivity of the Mapper graph.

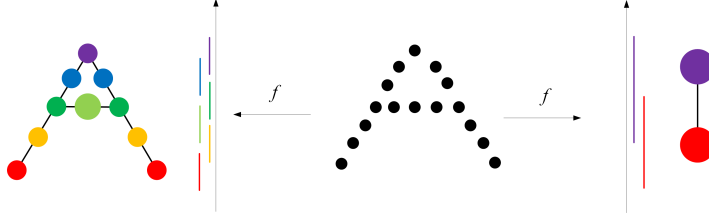


Figure 2: Sensitivity of the Mapper algorithm to resolution parameters: Both graphs use the same filter function (vertical axis projection). The left figure shows the result with $N = 7$ intervals and $p = 40\%$ overlap, while the right figure uses $N = 2$ intervals and $p = 25\%$ overlap.

4.1 Resolution Parameters Problem

In graph construction, Mapper is heavily influenced by parameter selection, and one of the most important parameters is the resolution parameter. The resolution parameter includes the number of intervals and the overlap rate; these two parameters are particularly important because they directly affect the output graph of the Mapper. Therefore, choosing these two parameters appropriately is crucial, as the resulting graph will show significant changes even with the smallest changes in input parameter selection. This is most clearly illustrated by the Figure 2

This is a major limitation of Mapper, and the problem is how we can automatically select parameters to ensure cluster quality while preserving the structure and interpretability of the resulting graph. This is a multi-objective optimization problem, requiring a balance between detailed coverage, noise immunity, and visual clarity. This problem can be solved as a black-box optimization problem on the resolution parameter space (N, p) , where N represents the number of coverage intervals and p is the overlap ratio, with the goal of determining configurations that maximize the quality of the Mapper graph according to a chosen evaluation criterion, such as cluster quality or topological signal preservation, without relying on manual adjustments or expert experience. This is a problem that needs to be solved in order to further expand and apply Mapper in practice.

4.2 Topology-Aware Black-Box Optimization Formulation

Instead of manually adjusting parameters, we propose S-Mapper as a topologically-oriented optimization process where population-based optimization algorithms such as PSO and GWO are used to automatically find suitable resolution parameter sets.

Mapper with resolution parameter optimization using PSO: **Input:** Dataset $X \subset \mathbb{R}^d$, lens function $f : X \rightarrow \mathbb{R}$; clustering algorithm $\mathcal{C}$.

Output: Optimized Mapper graph G^* .

Step 1: Compute the lens values

$$f(X) = \{f(x) : x \in X\}.$$

Step 2: Initialize a swarm of M particles, where each particle represents a candidate resolution pair (N_i, p_i) .

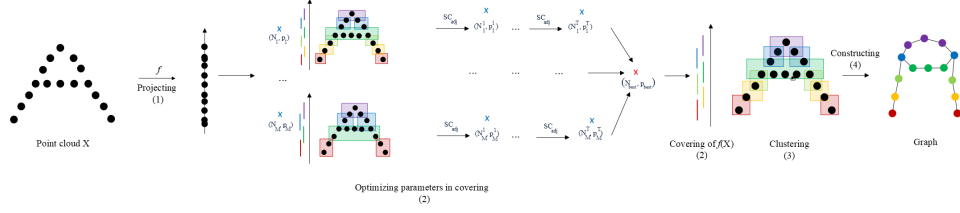


Figure 3: Step-by-step illustration of the S-Mapper algorithm applied to an A-shaped point cloud.

Step 3: Optimize the resolution parameters using PSO.

for $t = 1$ to T **do**

for each particle (N_i, p_i) **do**

 Round N_i to the nearest integer;

 Construct the Mapper graph $G_i = \text{Mapper}(X, f, \mathcal{C}, N_i, p_i)$;

 Evaluate G_i using $\text{TSC}(G_i)$;

 Update the personal and global best solutions;

end for

 Update particle velocities and positions;

end for

Step 4: Let $(N_{\text{best}}, p_{\text{best}})$ be the best resolution pair found by PSO.

Step 5: Construct the final optimized Mapper graph

$$G^* = \text{Mapper}(X, f, \mathcal{C}, N_{\text{best}}, p_{\text{best}}).$$

return G^* and $(N_{\text{best}}, p_{\text{best}})$.

In this process, each pair of parameters (N_i, p_i) is considered a randomly generated candidate. The positions of these candidates are continuously updated through each iteration according to the objective function TSC. This process unfolds as follows:

- **Projecting:** The high-dimensional dataset $X \subset \mathbb{R}^d$ is projected onto a one-dimensional space via a lens function $f : X \rightarrow \mathbb{R}$. A swarm of candidate resolution parameter pairs (N_i, p_i) is initialized, where N_i is the number of intervals and p_i is the overlap percentage.
- **Constructing cover on filter space:**
 - (2.1) *Optimizing parameters:* For each candidate (N_i, p_i) in the swarm: a cover of the projected data $f(X)$ is generated using N_i intervals and overlap p_i ; each interval is lifted back to the data space via the preimage $f^{-1}(I)$; a clustering algorithm is applied to the data in each preimage and resulting clusters are treated as nodes; edges are added between nodes whose clusters share at least one data point; the resulting Mapper graph is evaluated using the TSC score; and based on the optimization strategy (PSO or GWO), the swarm is updated to explore the parameter space effectively.
 - (2.2) *Covering filter space:* After reaching the convergence point, the final covering layer is constructed from the parameters found $(N_{\text{best}}, p_{\text{best}})$.

- **Clustering:** Apply the chosen clustering algorithm within the optimized coverage areas to partition the points.
- **Constructing Mapper graph:** A graph mapper is constructed with each node corresponding to a cluster, and nodes are connected by an edge if they share at least one data point.

The entire process automatically adjusts resolution settings to suit the data structure, promoting the creation of more stable and understandable graph mappers. The complete pipeline for this optimization-enhanced Mapper construction, S-Mapper, is illustrated step-by-step in Figure 3.

Mapper with resolution parameter optimization using GWO: **Input:** Dataset $X \subset \mathbb{R}^d$; lens function $f : X \rightarrow \mathbb{R}$; clustering algorithm $\mathcal{C}$.

Output: Optimized Mapper graph G^* .

Step 1: Compute the lens values

$$f(X) = \{f(x) : x \in X\}.$$

Step 2: Initialize a population of M wolves, where each wolf represents a candidate resolution pair (N_i, p_i) .

Step 3: Optimize the resolution parameters using GWO.

for $t = 1$ to T **do**

for each wolf (N_i, p_i) **do**

 Round N_i to the nearest integer;

 Construct the Mapper graph $G_i = \text{Mapper}(X, f, \mathcal{C}, N_i, p_i)$;

 Evaluate G_i using $\text{TSC}(G_i)$;

end for

 Update the α , β , and δ wolves;

 Update the remaining wolves according to the GWO update rules;

end for

Step 4: Let $(N_{\text{best}}, p_{\text{best}})$ be the best resolution pair obtained from the α wolf.

Step 5: Construct the final optimized Mapper graph

$$G^* = \text{Mapper}(X, f, \mathcal{C}, N_{\text{best}}, p_{\text{best}}).$$

return G^* and $(N_{\text{best}}, p_{\text{best}})$.

The general formulation described above can be implemented with different black-box optimization strategies for searching over the Mapper resolution-parameter space. In this work, the S-Mapper framework using two representative swarm-based metaheuristics: PSO and GWO. The purpose is not to exhaustively evaluate all possible optimizers, but to demonstrate that the proposed topology-aware formulation can be realized under two mechanistically different swarm search mechanisms. The concrete implementations of these strategies are detailed in Algorithms 1 and 2, which outline how candidate resolution parameters are explored, evaluated, and refined within each method. To better illustrate the role of resolution parameter optimization in Mapper construction, Figure 4 presents a comparison between the Mapper algorithm and the proposed S-Mapper framework

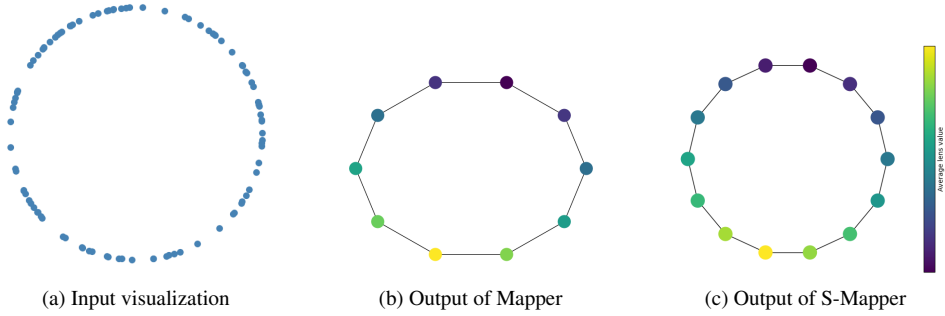


Figure 4: Illustration of the S-Mapper framework versus Mapper on a circular dataset.

on a simple circular dataset. Subfigure 4a shows the original data distribution in 2D space, exhibiting an evident circular structure. Subfigure 4b displays the output of the Mapper algorithm using manually selected parameters, while subfigure 4c presents the result of S-Mapper using swarm-based optimization to identify suitable resolution settings automatically. As shown, S-Mapper preserves the global shape more faithfully and yields a more coherent topological summary without requiring user parameter tuning.

5 Experiments

This section will clearly present the configurations and experimental environments as well as the experimental results achieved on the datasets in order to provide an objective effectiveness assessment of the robustness of S-Mapper. At the same time, compare S-Mapper with other parameter selection strategies.

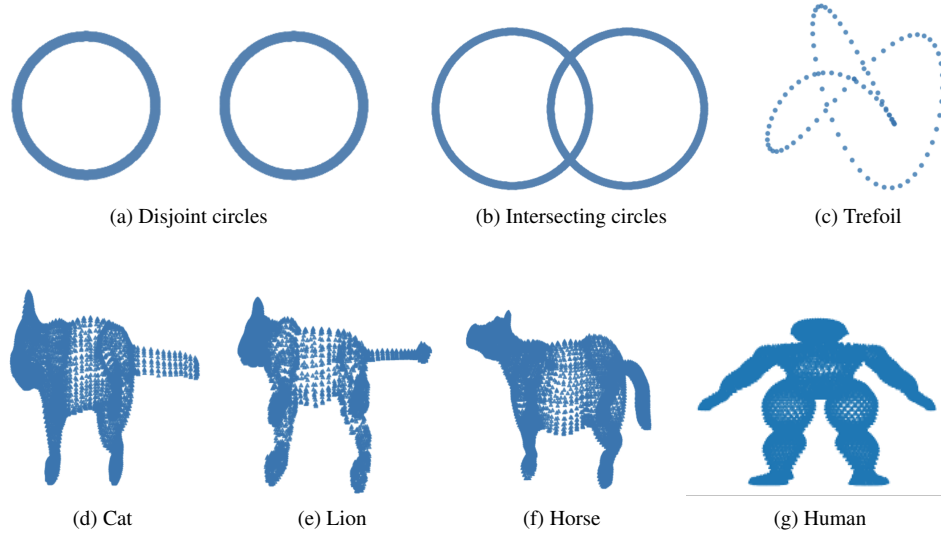
5.1 Datasets and Configuration

To comprehensively evaluate the effectiveness of the proposed S-Mapper framework, we conducted experiments on 9 datasets including 3 different types: aggregate data, 3D grid data and actual data. Datasets are selected by us to be used for many topological structures and different application contexts. The numerical datasets include two planar point clouds the disjoint and intersecting circles obtained from the Kepler Mapper library van Veen et al. (2019). The mesh datasets consist of 3D surface models, including cat, lion, horse Sumner and Popović (2004), and human Chen et al. (2009), which are widely used in 3D segmentation and deformation analysis. Figure 5 presents the 2D and 3D visualizations of the synthetic numerical and 3D mesh datasets. The real-world datasets include the diabetes dataset, initially published by Reaven and Miller Reaven and Miller (1979), and a COVID-19 RNA sequence dataset from GenBase, hosted by the National Genomics Data Center¹. For the COVID dataset, 3-mer ($k = 3$) frequency vectors were computed for each genome sequence, resulting in 64-dimensional feature vectors. Pairwise distances were calculated using Euclidean distance and normalized via min-max scaling, with the row-wise mean used as the filter function. A summary of all datasets is provided in Table 1.

The search space for resolution parameters was defined based on empirical knowledge of Mapper behavior: the overlap percentage is bounded to $p \in [0.1, 0.7]$ across all datasets,

Table 1 Summary of synthetic and real-world datasets used in the experiments, including key characteristics such as dimensionality and number of data points.

Characteristic	Disjoint circles	Intersecting circles	Trefoil	Cat	Lion	Horse	Human	Diabetes	COVID-19
Dimension	2	2	3	3	3	3	3	6	64
Number of data points	10000	10000	100	7207	4999	8430	4706	145	357

**Figure 5:** 2D and 3D visualizations of the synthetic numerical and 3D mesh datasets.

and the number of intervals N is bounded within $[N_{lb}, 25]$, where the lower bound is dataset-dependent. The specific bounds used for each dataset are summarized in Table 2. These limits were chosen based on our experience in graph creation, and we estimated that within that range we would obtain a well-interpreted Mapper graph. These limits could be extended, but this would require additional resources.

Table 2 Search space bounds for resolution parameters across datasets.

Parameter	Disjoint circles	Intersecting circles	Trefoil	Cat	Lion	Horse	Human	Diabetes	COVID-19
N range	[5, 25]	[5, 25]	[5, 25]	[7, 25]	[9, 25]	[5, 25]	[7, 25]	[5, 25]	[10, 20]
p range	[0.1, 0.7]	[0.1, 0.7]	[0.1, 0.7]	[0.1, 0.7]	[0.1, 0.7]	[0.1, 0.7]	[0.1, 0.7]	[0.1, 0.7]	[0.1, 0.7]

All experiments were implemented in Python and executed on an Apple MacBook Air with M3 chip and 16GB RAM. The Mapper algorithm was implemented using the Kepler Mapper library van Veen et al. (2019), with fixed random seeds employed during graph construction and clustering. The source code is publicly available at ². The Mealpy library³ was used to implement both PSO and GWO, with population size and number of iterations set to 20, and random seed fixed at 42 Van Thieu and Mirjalili (2023). For comparison, D-Mapper was evaluated using the official implementation released by its authors Tao and Ge (2025a). The parameter configurations of Mapper and D-Mapper that we selected are

presented in Tables 3 and 4. The parameters were adopted from the original paper and further adjusted based on preliminary experiments.

Table 3 Mapper resolution parameters (N and p) for each dataset used in the experiments.

Parameter	Disjoint circles	Intersecting circles	Trefoil	Cat	Lion	Horse	Human	Diabetes	COVID-19
N	19	20	6	9	12	12	8	5	13
p	10%	50%	40%	20%	20%	4%	5%	30%	10%

Table 4 D-Mapper resolution parameters (N and α) for each dataset used in the experiments.

Parameter	Disjoint circles	Intersecting circles	Trefoil	Cat	Lion	Horse	Human	Diabetes	COVID-19
N	13	8	7	9	9	11	7	4	15
α	0.120	0.088	0.030	0.010	0.050	0.030	0.010	0.400	0.006

5.2 Results and Analysis

To assess the performance of the proposed S-Mapper framework, we conducted a series of comparative evaluations against two baseline approaches: the Mapper algorithm and the distribution guided D-Mapper. To isolate the contribution of the swarm-based search itself, we additionally include Random Mapper as an ablation baseline, in which resolution parameters are drawn uniformly at random within the same search bounds; the best-scoring random configuration is retained. This baseline directly addresses whether the structured search machinery of PSO and GWO is necessary at all, or whether comparable resolution parameters could be found by uninformed sampling of the same space. For completeness, Table 5 summarizes the final resolution parameters selected by S-Mapper using PSO and GWO. These optimized configurations are used to construct the S-Mapper graphs evaluated in the following comparative analysis. These evaluations were conducted across nine benchmark datasets, including synthetic numerical point clouds, 3D meshes, and real-world scientific data. The main evaluation metrics we used in this study include NSC for assessing cluster quality, TSR for assessing structural integrity, and finally TSC for a comprehensive assessment of both graph quality and topological accuracy.

Figures from 6–14 show graph results created by Mapper, Random Mapper, D-Mapper and S-Mapper on nine datasets. Graphs created by S-mapper ensure structural coherence and topological characteristics are clearer on most datasets. On the graph, clusters are clearly displayed separately and the ability to preserve the geometric structure is better than other strategies. This completeness is shown by complex datasets such as Cat, Lion, Horse, in which Mapper, Random Mapper and D-Mapper tend to create overly fragmented topological structures or inefficient organization. In addition, the graphs obtained by D-Mapper show moderate improvements compared to Mapper, especially in datasets with uneven distribution. D-Mapper can interpret features that more closely match the data structure; however, D-Mapper is less consistent and less stable than S-Mapper. By using PSO and GWO to optimize resolution parameters, S-Mapper can produce more stable graphs with stronger topology across multiple datasets, while ensuring a performance balance between clustering quality and topology preservation.

When applied to a real-world diabetes dataset, S-Mapper demonstrates superior ability to preserve core insights into the topological structure first identified by Reaven and Miller using projection search. Their analysis revealed a central cluster with two distinct branches corresponding to diabetes onset in adults and childhood. The results obtained from the S-Mapper graph show that it can easily capture these branching structures and interpret them better than other methods where the graph representation remains unclear. This demonstrates the value of the S-Mapper not only as a structural summarization tool but also as a practical discovery tool for identifying features. In the COVID-19 dataset, the graph generated by the S-Mapper can reflect the evolutionary divergence of viral lineages, showing the process of vertical evolution. Furthermore, groups with distinct structures offer biological or epidemiological implications; however, this requires further research. While the S-Mapper dataset highlights subtle lineage branching and demonstrates its potential in epidemiology, it still has limitations compared to the interpretive capabilities of the D-Mapper. While the S-Mapper only shows vertical evolution, the D-Mapper represents both vertical and horizontal evolution thanks to its mechanism of closely following the data’s shape structure. This result is due to the fact that the D-Mapper has the ability to self-adjust to closely follow the data structure, while the original Mapper cannot do that, leading to the parameter optimization on the S-Mapper also being partially affected. Although D-Mapper retains more biological detail in this case, it requires complex assumptions about data distribution or constituent probability structure. Conversely, S-Mapper, as a black-box optimizer, does not require knowledge of the data’s hidden distribution, making it easier to implement and more generalizable to unstructured data.

Table 5 Optimized resolution parameters selected by S-Mapper using PSO and GWO across all benchmark datasets.

Dataset	PSO		GWO	
	N^*	p^*	N^*	p^*
Disjoint circles	18	0.100	23	0.100
Intersecting circles	25	0.120	5	0.100
Trefoil	6	0.174	6	0.174
Cat	7	0.100	7	0.100
Lion	9	0.100	9	0.100
Horse	5	0.100	5	0.100
Human	7	0.100	7	0.100
Diabetes	5	0.454	5	0.455
COVID-19	10	0.121	10	0.121

In addition, we also conducted a comparison based on the quantitative results presented in Table 6. As observed, both S_{PSO} -Mapper and S_{GWO} -Mapper outperformed Mapper, Random Mapper, and D-Mapper on most datasets across all three indices: NSC, TSR, and TSC. However, the results obtained by D-Mapper (0.820) are slightly better than those obtained by S-Mapper (0.805) in the dataset of two intersecting circles. This difference is due to the fact that the overlay construction of D-Mapper takes into account the distribution of the data, thus helping it to closely follow the geometric structure. And partly due to the nature of spatial partitioning in traditional Mappers (which S-Mapper inherits), it’s impossible to create dynamic partitions like how D-Mappers build covers. Conversely, on the dataset of two discrete circles, S-Mapper achieved superior results with 0.835, thus demonstrating

Table 6 Evaluation comparison of the Mapper, Random Mapper, D-Mapper, and S-Mapper algorithms—optimized via PSO and GWO—across all benchmark datasets.

Dataset	Mapper			Random Mapper			D-Mapper			S _{PSO} -Mapper			S _{GWO} -Mapper		
	NSC	TSR	TSC	NSC	TSR	TSC	NSC	TSR	TSC	NSC	TSR	TSC	NSC	TSR	TSC
Disjoint circles	0.651	1.000	0.825	0.655	1.000	0.827	0.637	1.000	0.818	0.669	1.000	0.835	0.669	1.000	0.835
Intersecting circles	0.451	1.000	0.725	0.594	1.000	0.797	0.639	1.000	0.820	0.610	1.000	0.805	0.610	1.000	0.805
Trefoil	0.482	0.500	0.491	0.562	0.500	0.531	0.530	0.250	0.390	0.572	0.500	0.536	0.572	0.500	0.536
Cat	0.499	1.000	0.750	0.527	1.000	0.764	0.511	1.000	0.756	0.590	1.000	0.795	0.590	1.000	0.795
Lion	0.514	0.500	0.507	0.513	1.000	0.757	0.527	0.667	0.597	0.563	1.000	0.782	0.563	1.000	0.782
Horse	0.590	0.500	0.545	0.579	1.000	0.790	0.469	0.800	0.634	0.661	1.000	0.830	0.661	1.000	0.830
Human	0.619	1.000	0.809	0.580	1.000	0.790	0.486	0.500	0.493	0.642	1.000	0.821	0.642	1.000	0.821
Diabetes	0.459	0.222	0.340	0.468	0.143	0.305	0.454	0.143	0.299	0.491	0.286	0.389	0.491	0.286	0.389
COVID-19	0.546	0.250	0.398	0.559	0.500	0.530	0.479	0.091	0.285	0.581	0.500	0.540	0.581	0.500	0.540

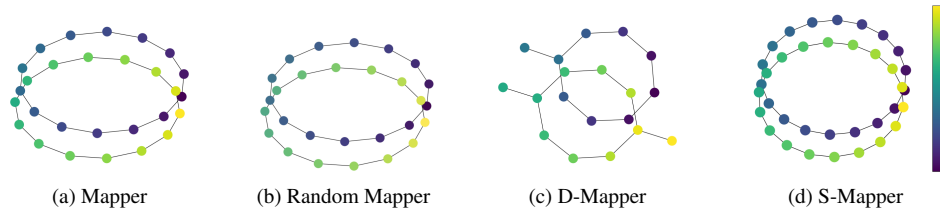


Figure 6: Comparison of Mapper, Random Mapper, D-Mapper, and S-Mapper outputs on the disjoint circles dataset.

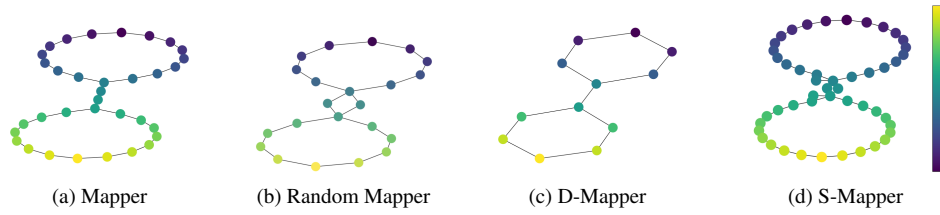


Figure 7: Comparison of Mapper, Random Mapper, D-Mapper, and S-Mapper outputs on the intersecting circles dataset.

its ability to represent cluster quality as well as preserve structure. Similarly, for complex datasets such as Horse and Human, the variants of S-Mapper also obtained higher results of 0.830 and 0.821 respectively when directly compared with Mapper and D-Mapper. Moving on to real-world datasets like diabetes and COVID-19, S-Mapper achieves competitive performance on the real-world datasets. It not only produces high-quality topologies but also maintains consistent TSR values across datasets, highlighting its effectiveness in creating stable and easily understandable graphs. Besides, Random Mapper, which is a method that samples the resolution parameters uniformly within the same search range and with the same evaluation method, always shows lower performance than both S-Mapper variants on most

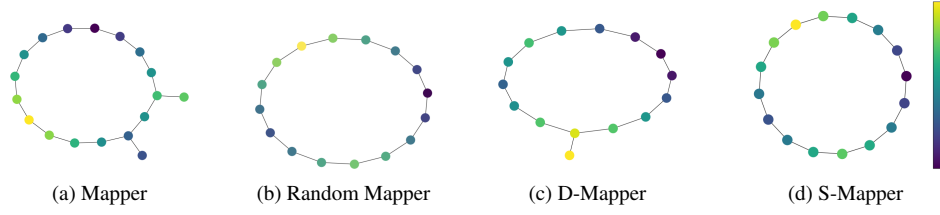


Figure 8: Comparison of Mapper, Random Mapper, D-Mapper, and S-Mapper outputs on the trefoil dataset.

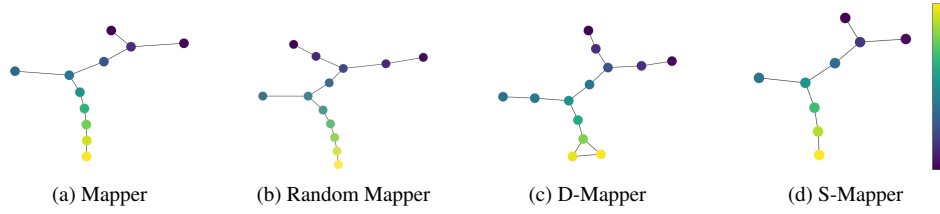


Figure 9: Comparison of Mapper, Random Mapper, D-Mapper, and S-Mapper outputs on the cat dataset.

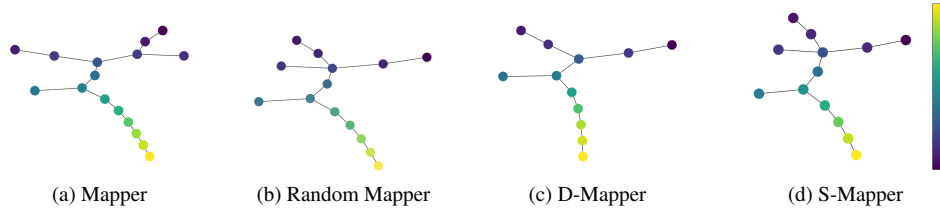


Figure 10: Comparison of Mapper, Random Mapper, D-Mapper, and S-Mapper outputs on the lion dataset.

datasets. Thereby, it proves that the performance improvements originate from the structured swarm search process, instead of merely evaluating more configurations with respect to the TSC objective function. Notably, on datasets such as Horse and Lion, Random Mapper achieves higher TSC than D-Mapper (0.790 vs. 0.634 and 0.757 vs. 0.597, respectively), suggesting that TSC-guided scoring alone provides a useful signal, while the swarm-based search further amplifies this advantage.

On the diabetes dataset, which showed two distinct patient groups, both the S_{PSO} -Mapper and the S_{GWO} -Mapper showed the highest TSC (0.389). Similarly, for the COVID-19 dataset, the S-Mapper achieved both the highest NSC and TSR of 0.581 and 0.500 respectively, demonstrating its ability to extract and retain data structure. The TSR score achieved on the D-Mapper was still low in the diabetes dataset (0.143) and COVID-19 dataset (0.091), indicating that the data structure recovery ability is still limited. In contrast to the swarm-based optimization strategy, the S-Mapper can achieve higher adaptability and explanatory

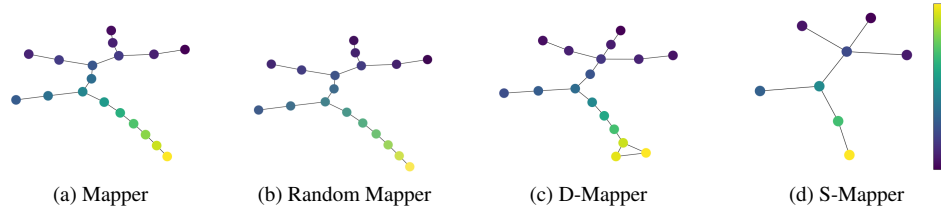


Figure 11: Comparison of Mapper, Random Mapper, D-Mapper, and S-Mapper outputs on the horse dataset.

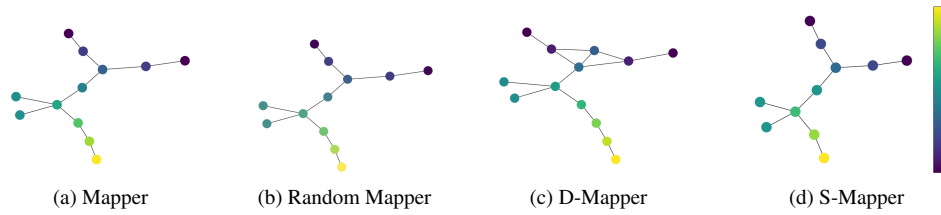


Figure 12: Comparison of Mapper, Random Mapper, D-Mapper, and S-Mapper outputs on the human dataset.

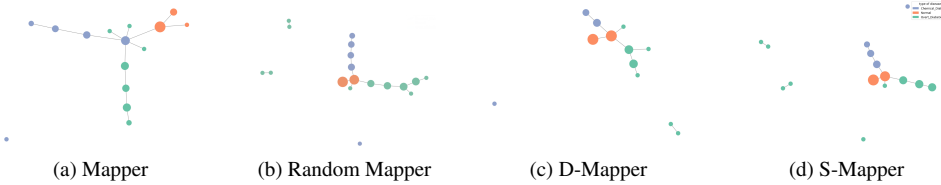


Figure 13: Comparison of Mapper, Random Mapper, D-Mapper, and S-Mapper outputs on the diabetes dataset.

power with fundamentally structured data. These advantages also reinforce the effectiveness of S-Mapper in creating stable and interpretably meaningful graphs.

Simultaneously, to verify the above results, we also conducted statistical probability tests using Wilcoxon based on the TSRs obtained from all datasets. The above results show that S_{PSO} -Mapper and the S_{GWO} -Mapper both show significantly superior results compared with Mapper, specifically $W = 45$, $p\text{-value} = 0.002$, and D-Mapper, reaching $W = 44$, $p\text{-value} = 0.004$, at the significance level with $\alpha = 0.05$. These findings statistically demonstrate that the performance improvements of S-Mapper are not due to random variation but accurately reflect improvements in the quality of the topological graph through swarm-based optimization of the resolution parameters.

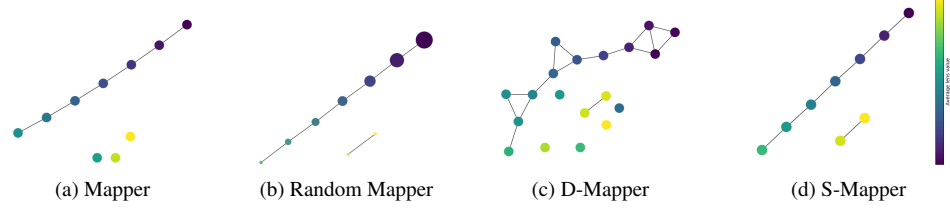


Figure 14: Comparison of Mapper, Random Mapper, D-Mapper, and S-Mapper outputs on the COVID-19 dataset.

5.3 Discussion

The near-identical final TSC scores achieved by PSO and GWO provide empirical support for the robustness of the proposed framework: the quality of the optimized Mapper graph appears not to be an artifact of a single optimizer’s search strategy, but rather a stable outcome under the TSC-guided optimization setting. This consistency across two mechanistically distinct optimizers further reinforces that robustness is a property of the framework itself rather than of any particular search algorithm. This optimizer-agnostic behavior should be interpreted as an observed property of S-Mapper rather than as redundancy between two interchangeable optimizers. Specifically, it shows that optimization algorithms based on different populations can be integrated into the same framework. The similarity in the obtained TSC values does not mean that the choice of optimizer is irrelevant; instead, it shifts the comparison from the quality obtained to the convergence behavior. To further analyze the efficiency and convergence behavior of each different optimization strategy, we compared the performance of S_{PSO} -Mapper and S_{GWO} -Mapper in terms of both the number of convergence cycles and the convergence time, as shown in Figures 15 and 16.

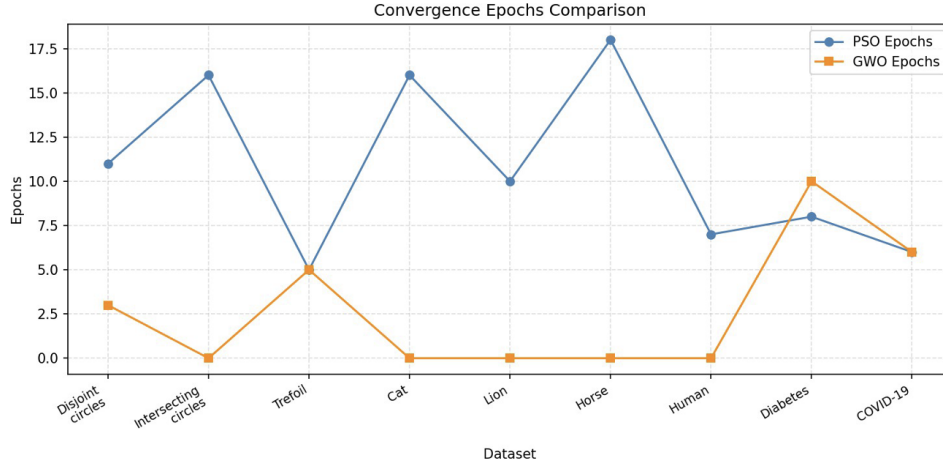


Figure 15: Performance comparison on the convergence epoch of S-Mapper using PSO and GWO optimization strategies across benchmark datasets.

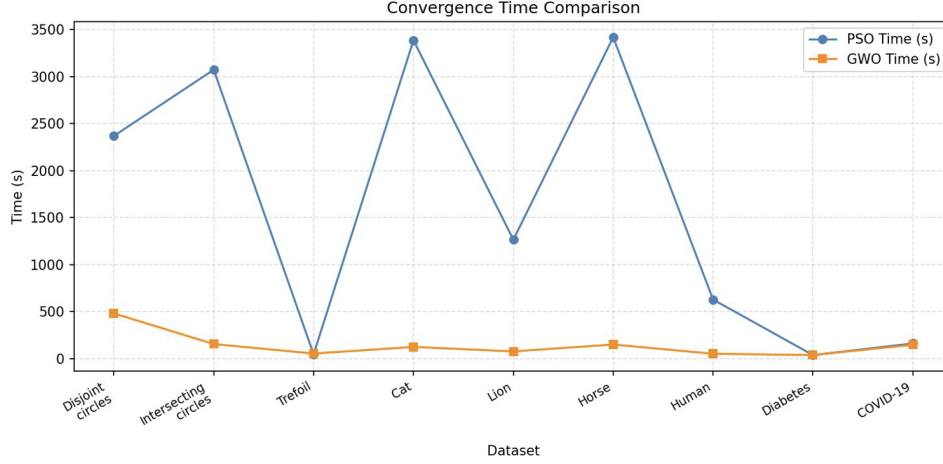


Figure 16: Performance comparison on the convergence time of S-Mapper using PSO and GWO optimization strategies across benchmark datasets.

Overall, the S_{GWO} -Mapper variant demonstrates superior convergence efficiency in most cases, often requiring significantly fewer optimization epochs than the S_{PSO} -Mapper. For example, on the disjoint circles and lion datasets, GWO converges in 3 and 0 epochs, respectively, compared to 11 and 10 epochs for PSO. This could be due to a random factor when initializing the candidate in the GWO, accidentally falling into a convergence point, thus causing the GWO to converge right from epoch 0. On the cat and horse datasets, $SGWO$ -Mapper reduces total optimization time from approximately 3380s and 3420s (under PSO) to approximately 140s and 160s, respectively representing nearly a 95% reduction in runtime. An exception is observed in the diabetes dataset, where PSO achieves convergence in fewer epochs 8 vs 10, attributable to the relatively small size and lower structural complexity of that dataset. We also conducted a sensitivity analysis to assess the stability of TSC when the weights w_1 and w_2 were changed on the Trefoil dataset. The results are clearly shown in Figure 17, which shows that the recovered topological structure maintained stability over a fairly wide range of values when all weight configurations revolved around the default setting. At the same time, the data structure was fully preserved. However, in extreme cases where only TSR ($w_1 = 0.0$) or NSC ($w_1 = 1.0$) were prioritized, the results began to change. Prioritizing only NSC cluster quality would cause the structures to break and split into multiple clusters, while relying solely on TSR, although preserving the original structure, would result in poor cluster quality.

To further assess the stability of the S-Mapper for the hyperparameters of the optimization algorithm, a sensitivity analysis is essential. On the Trefoil dataset, we used PSO for this. The results are shown in , which shows some variation in TSC across different iterations, but overall it remained stable around the best reported value of 0.536. Similarly, the TSC score is relatively stable across different population sizes, consistently hovering near the best reported value of 0.536.

Three failure modes are visible in the results. First, on datasets with few data points, the resolution search has little structure to exploit: the Trefoil dataset, with only 100 points, attains the lowest absolute TSC (0.536) and shows the smallest margin over the baselines. Second, on datasets where the intrinsic topological signal is weak, the TSR

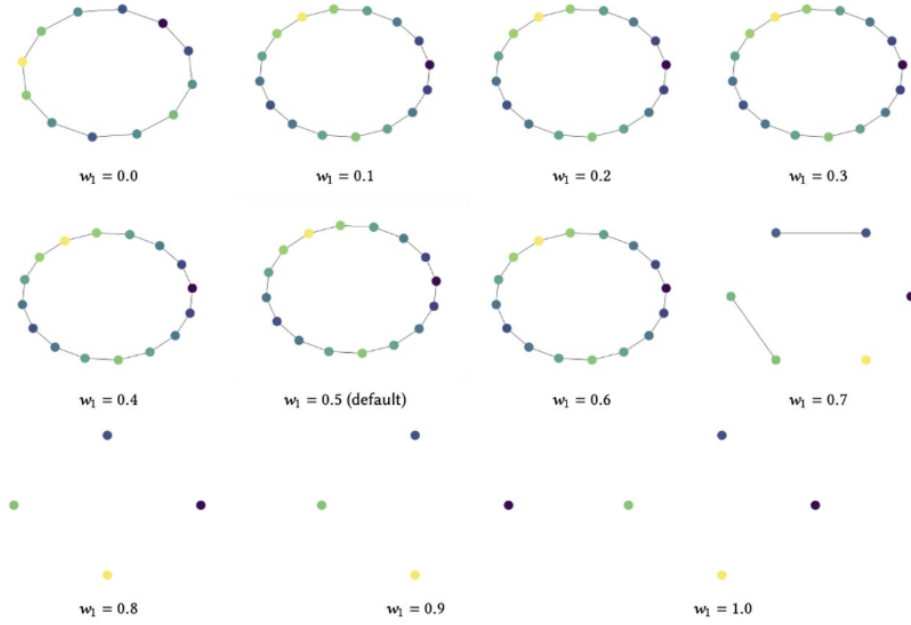


Figure 17: Weight sensitivity analysis on the Trefoil dataset. S-Mapper produces stable Mapper topologies across a broad range of weight configurations around the default setting ($w_1 = w_2 = 0.5$, boxed). Structural distortions become apparent at the extreme cases ($w_1 = 0.0$ and $w_1 = 1.0$).

component saturates at a low value and caps the achievable TSC regardless of how well the parameters are tuned (evident on Diabetes and COVID-19). Third, S-Mapper optimizes only the resolution parameters and takes the lens function as fixed, so an ill-suited lens bounds the quality of every configuration the swarm can reach.

Despite its strong empirical performance, S-Mapper has several limitations. First, it inherits Mapper’s dependence on the lens function. Second, the quality of the output remains sensitive to the choice of clustering algorithm. Third, the iterative evaluation of candidate configurations incurs notable optimization overhead. Fourth, the present study instantiates the framework only with two swarm-based optimizers; a systematic comparison with Bayesian optimization using TSC as the acquisition objective would further contextualize the advantage of population-based search. Fifth, S-Mapper is currently evaluated using TSC, which also serves as its optimization objective; future work should instantiate the framework with alternative topology-aware objectives such as graph entropy, modularity, or agreement with expected Betti numbers used directly as the optimization target, which would both provide external validation and empirically demonstrate the framework’s independence from any single quality signal, as well as multi-objective formulations that balance clustering quality and topological fidelity along a Pareto front rather than through a fixed scalarization.

6 Conclusion

This study presented S-Mapper, a general topology-aware optimization paradigm for adaptive Mapper construction. The proposed framework reformulates the selection of the number of intervals and overlap percentage as a black-box optimization problem guided by topological quality signals, thereby reducing reliance on heuristic or manual tuning. Importantly, the key contribution of S-Mapper lies in its topology-aware optimization formulation rather than in any specific optimizer or quality metric. TSC and the swarm optimizers (PSO, GWO) are interchangeable instantiations: TSC defines what topological quality means and can be replaced by alternative topology-aware objectives, while PSO and GWO define how the search is conducted and can be replaced by alternative black-box optimization strategies. By reducing Mapper’s dependence on user-defined parameters, such as the number of intervals and the overlap ratio, S-Mapper contributes to improving the stability as well as the interpretability of topological representations. Experimental results on both synthetic data and real-world data show that S-Mapper frequently achieves better performance than Mapper and D-Mapper in terms of clustering quality, topological fidelity and stability. In addition, the comparative analysis between two popular swarm optimization strategies, PSO and GWO, shows differences in convergence behavior, thereby further clarifying the practical trade-offs when applying them in TDA.

This work lays the groundwork for scalable, data-driven Mapper pipelines by reducing reliance on manual parameter tuning. Beyond empirical improvements, S-Mapper contributes to the development of automated topological simplification and graph construction. A potential development direction is to continue strengthening the mathematical foundation of S-Mapper through connecting the structure of this method with Morse theory. This connection may contribute to further clarifying the relationship between discrete Mapper outputs and the corresponding topological features. Besides, future studies may extend the proposed framework to multi-lens Mapper, consider other black-box optimization strategies beyond SI, as well as deploy the method on high-dimensional biomedical and network datasets, where topological stability and sensitivity to the resolution parameter remain important issues. It can also be expanded to optimize other input parameters such as lens and clustering.

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Declaration of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The datasets and source code used in this study are publicly available through the references and repositories cited in the manuscript. Additional materials are available from the corresponding author upon reasonable request.

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